

Singular Value Decomposition

Geometric Algorithms

Objectives

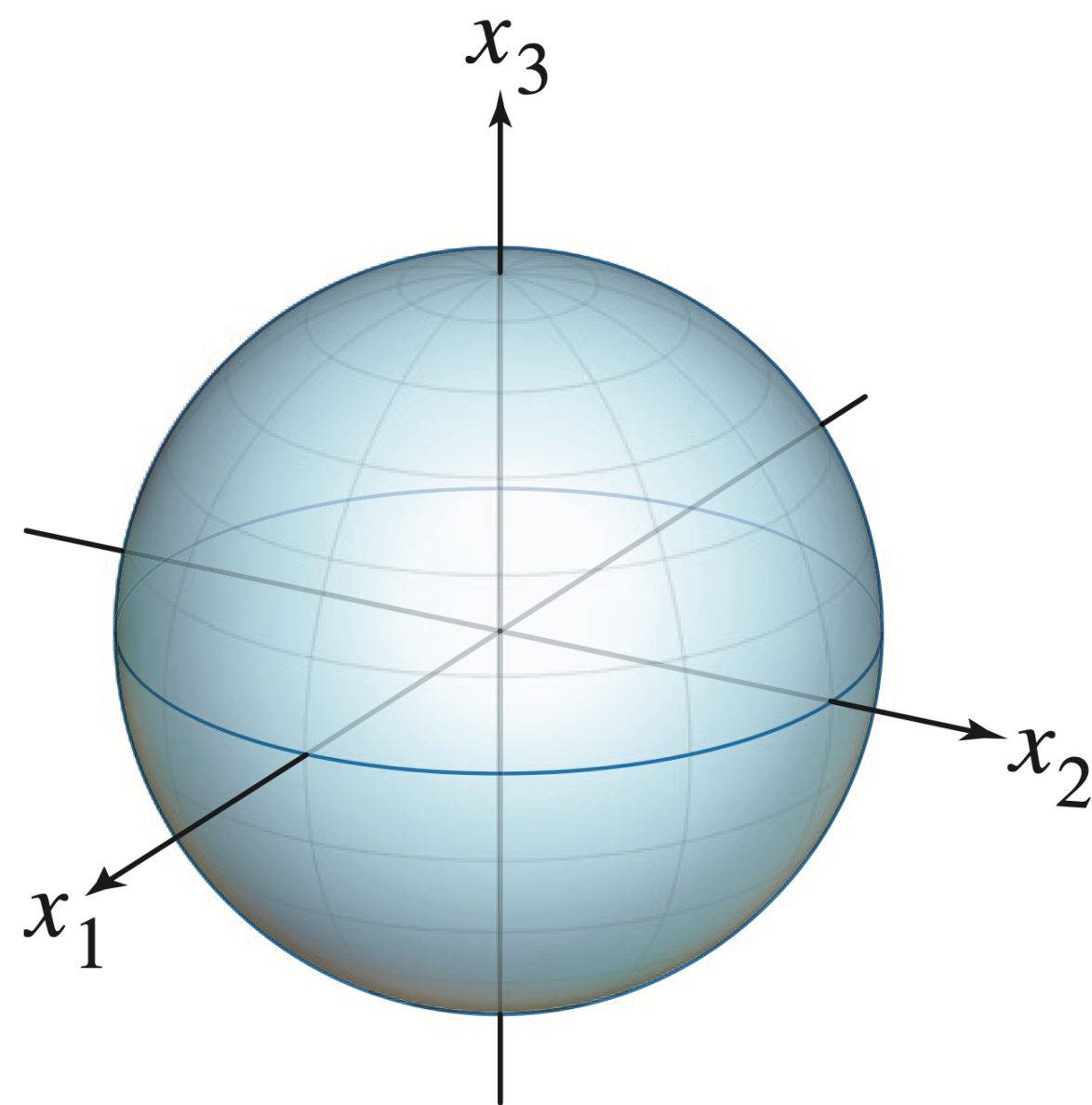
Introduce the **singular value decomposition**

(probably the most important decomposition for CS)

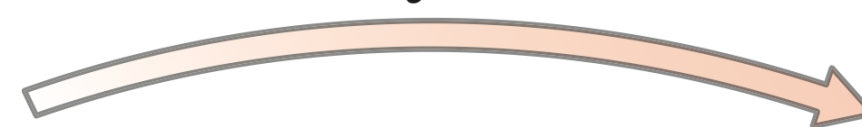
Motivation

Question

What shape is a the unit sphere after a linear transformation?



Multiplication
by A

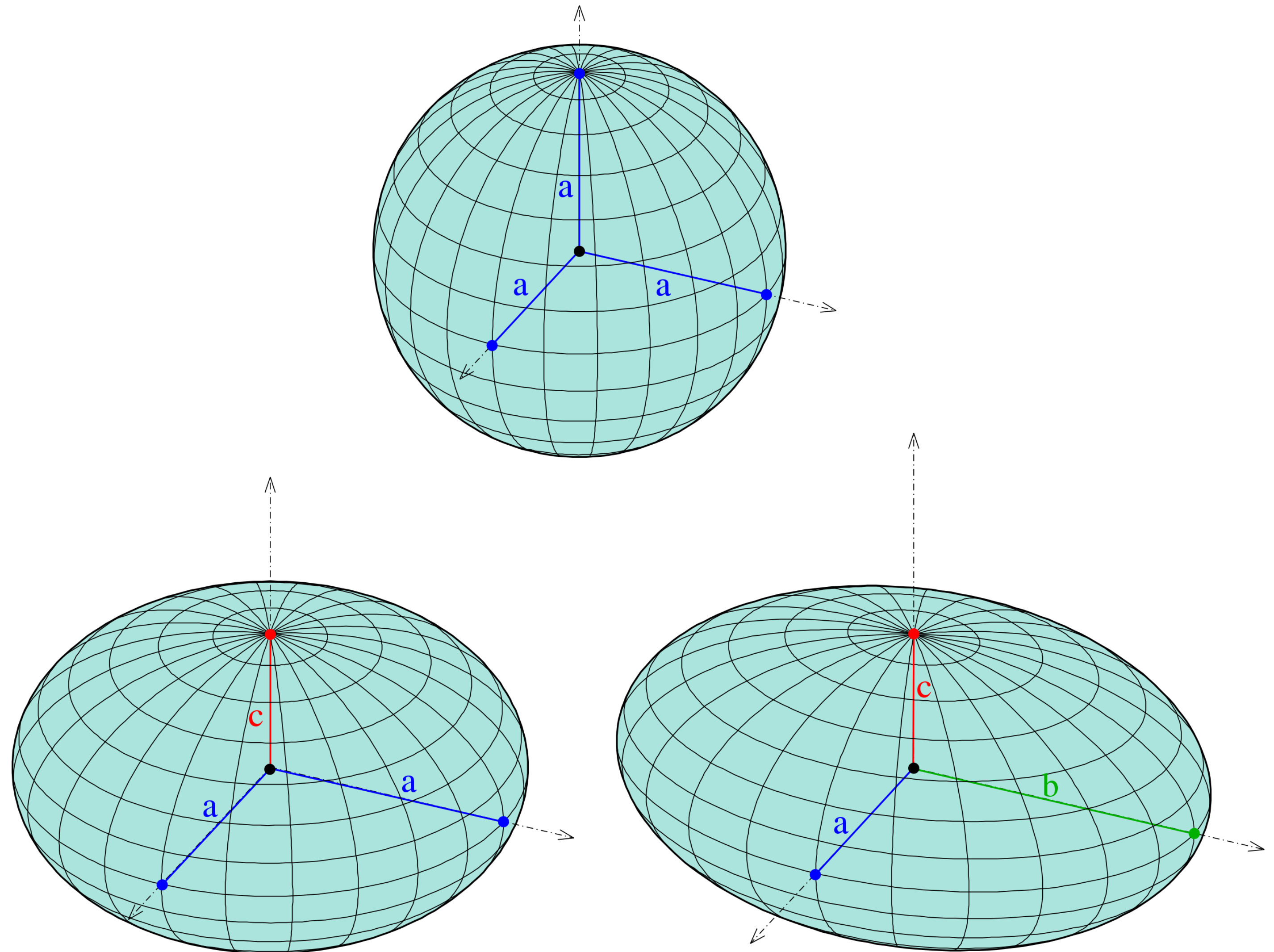


???

Ellipsoids

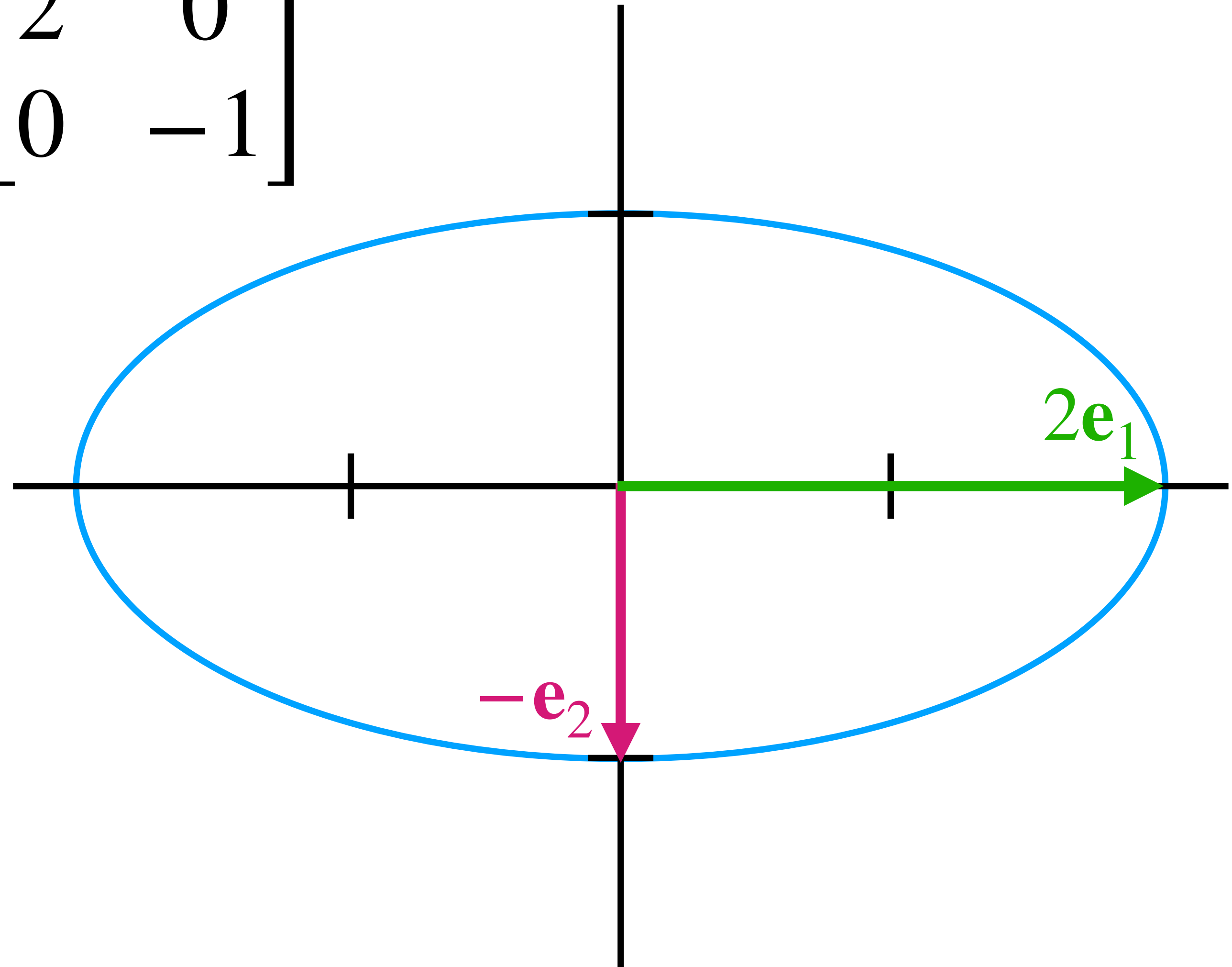
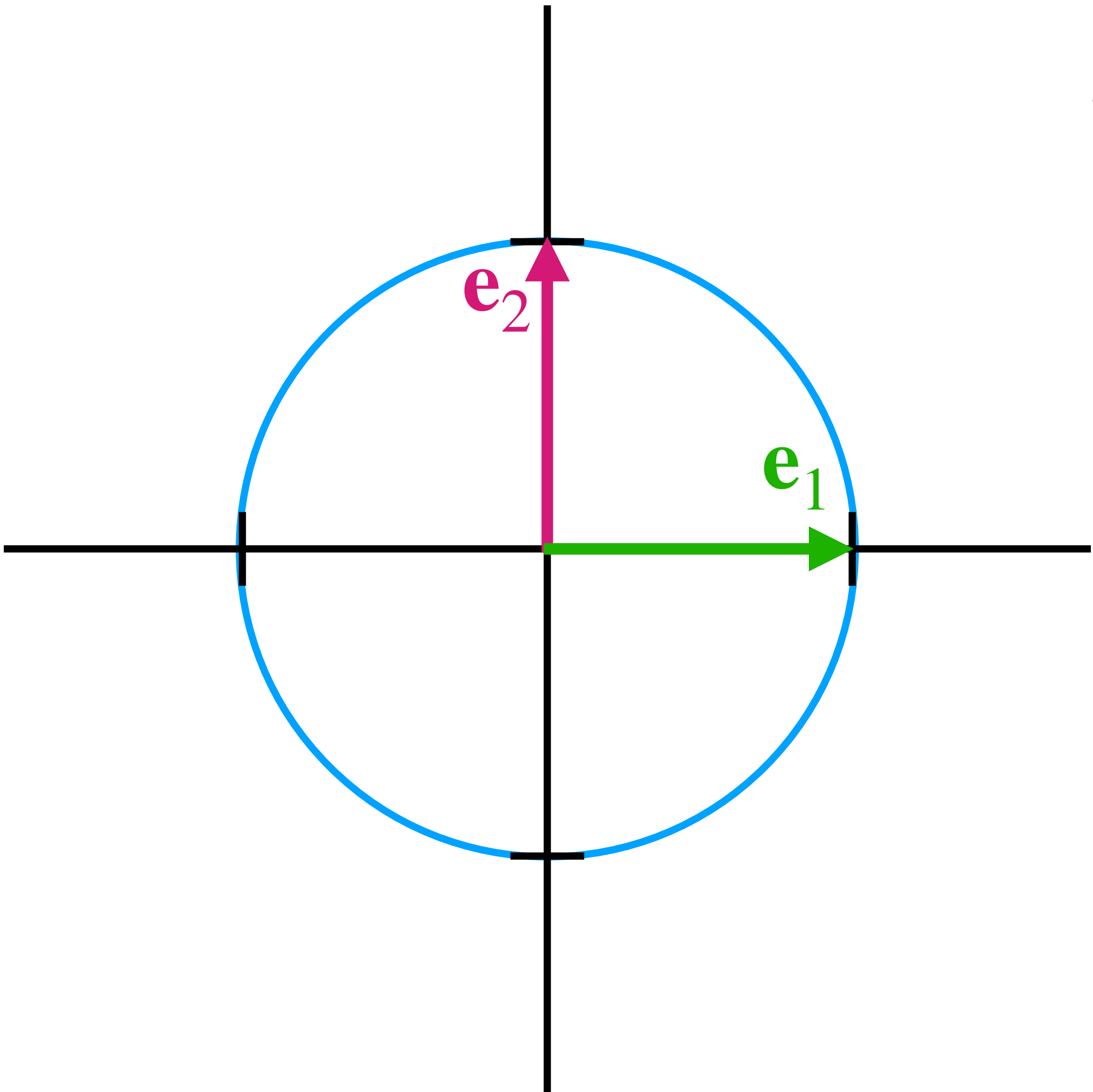
Ellipsoids are spheres
"stretched" in
orthogonal directions
called the
axes of symmetry or the
principle axes

Linear trans. map
spheres to ellipsoids

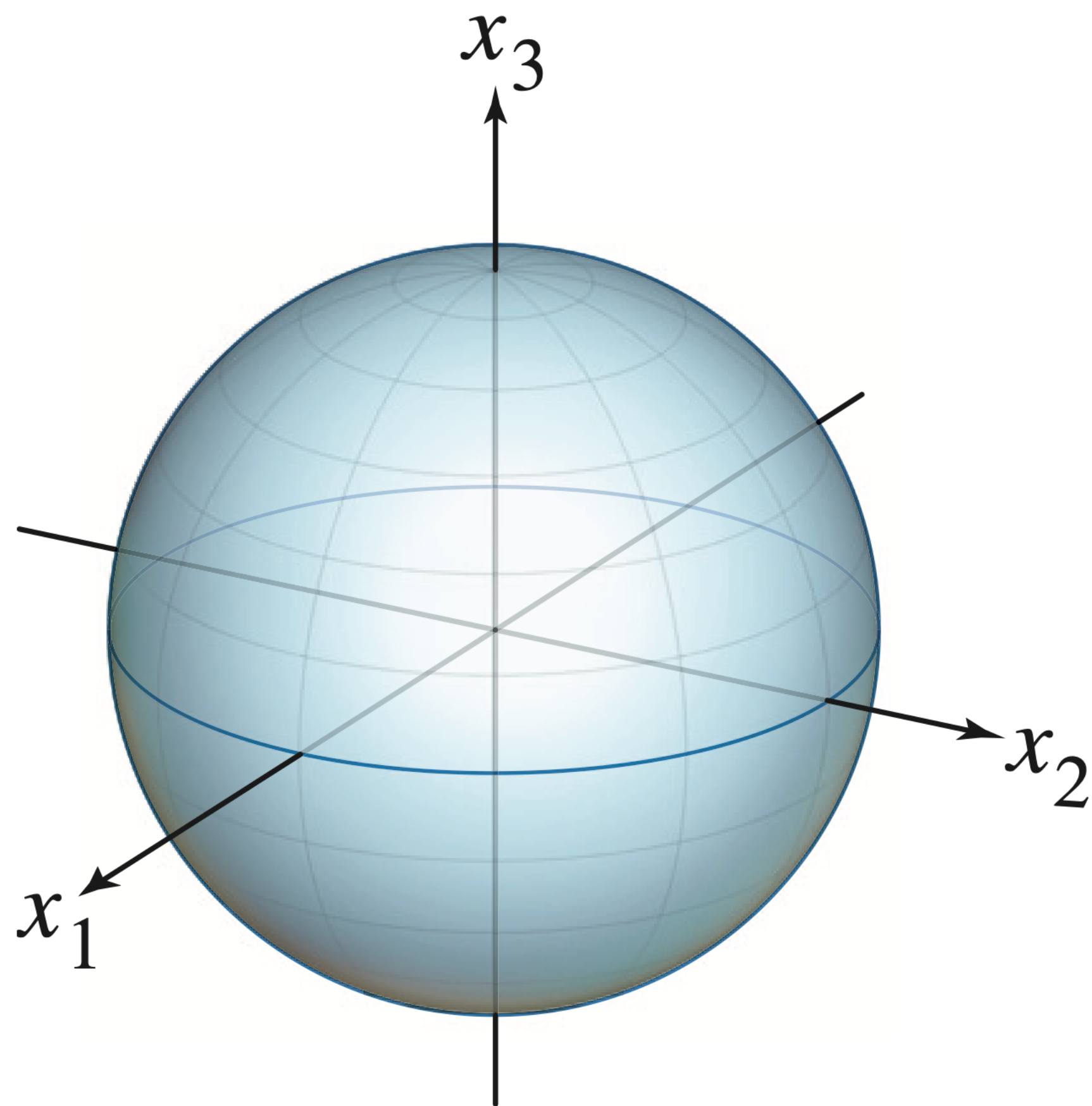


Simple Example : Scaling Matrices

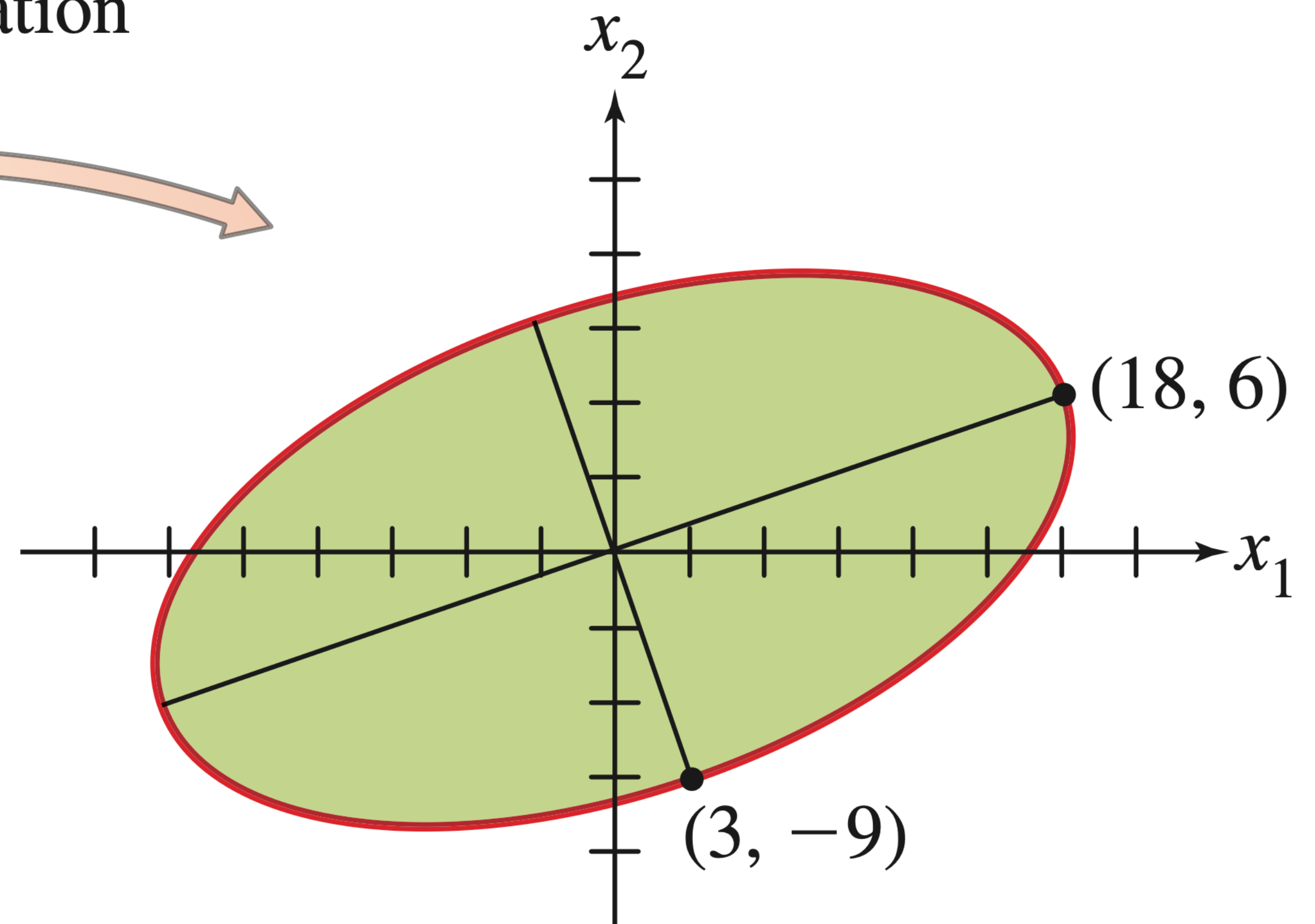
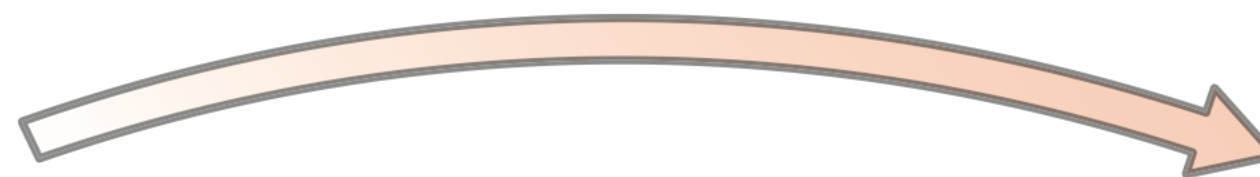
$$A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$$



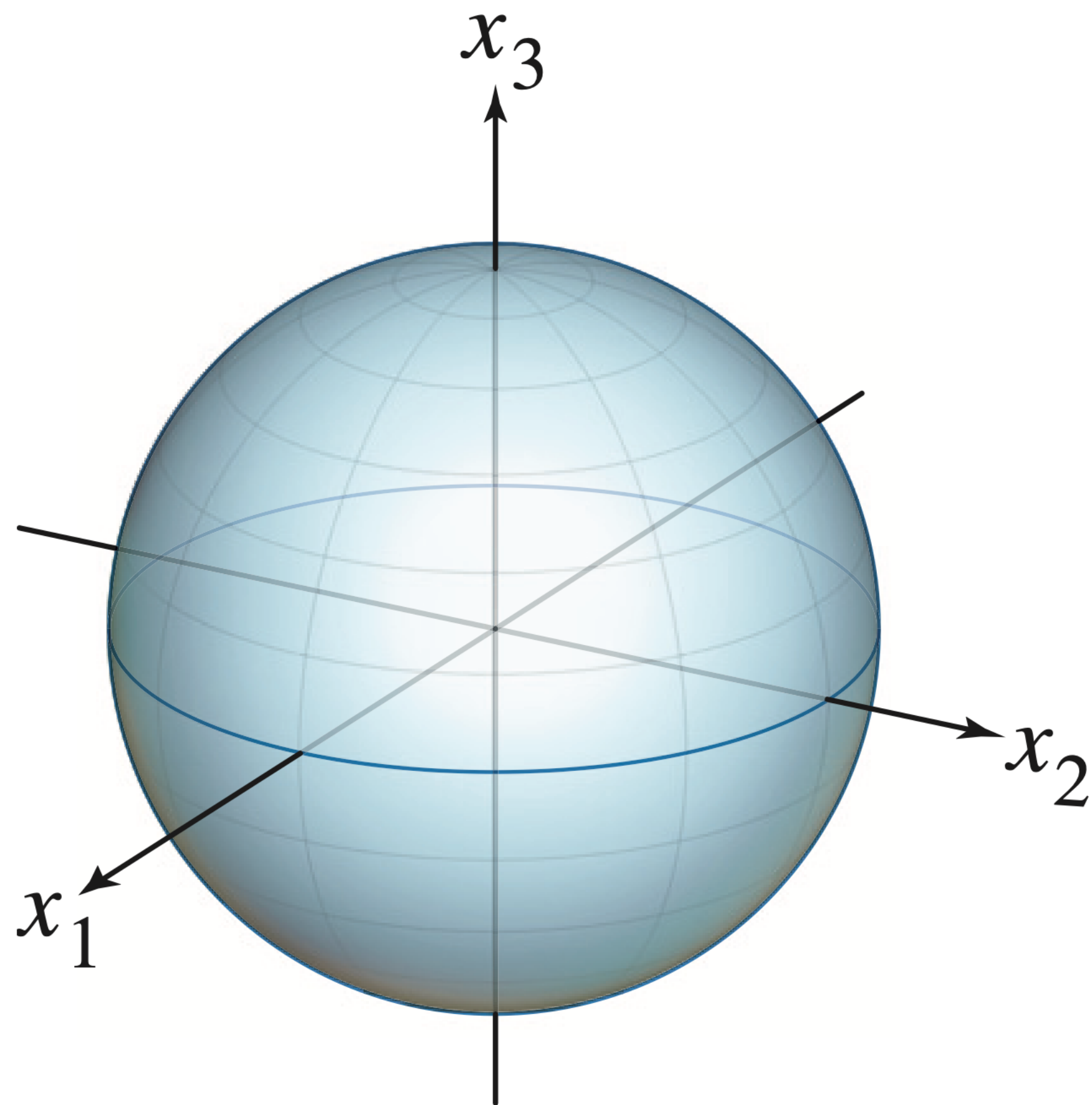
The Picture



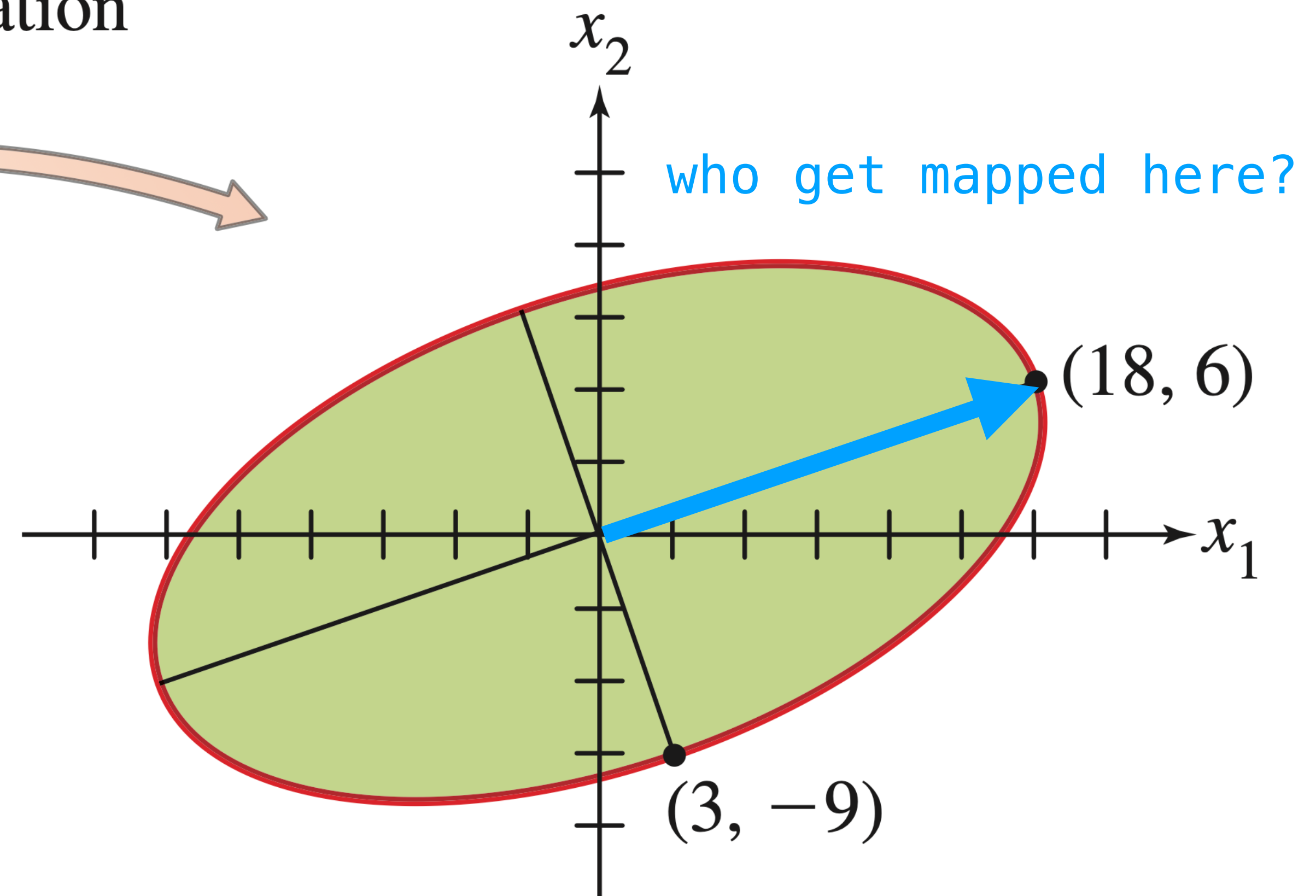
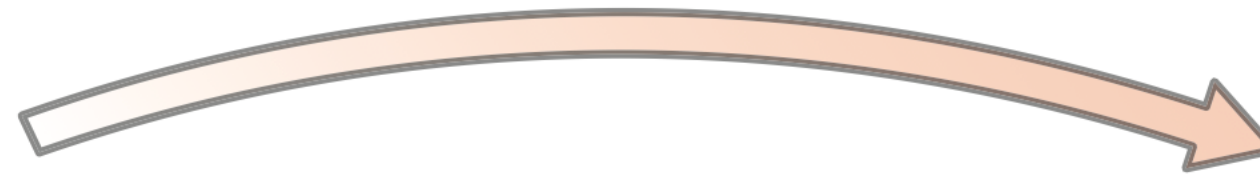
Multiplication
by A



The Picture

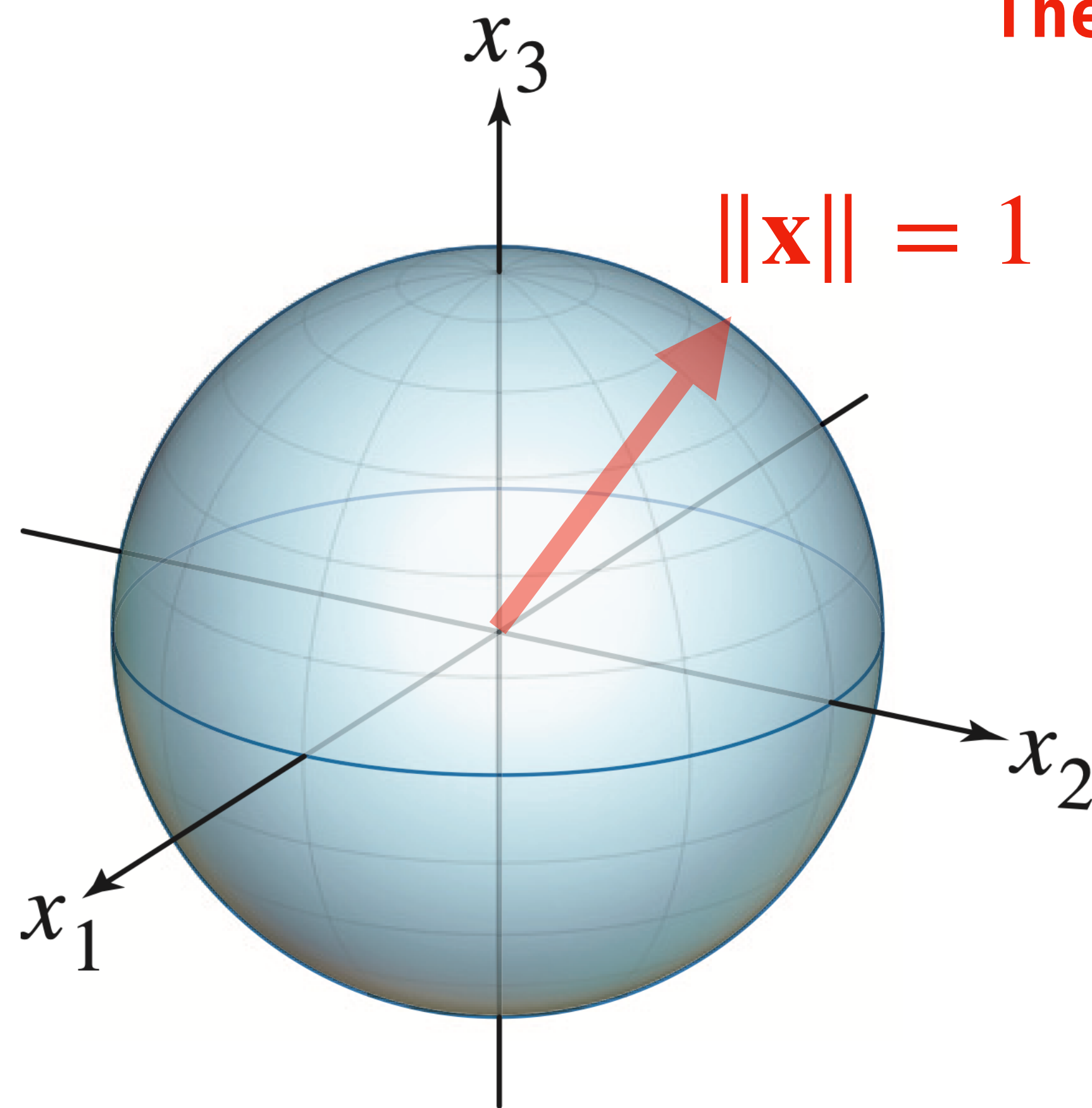


Multiplication
by A

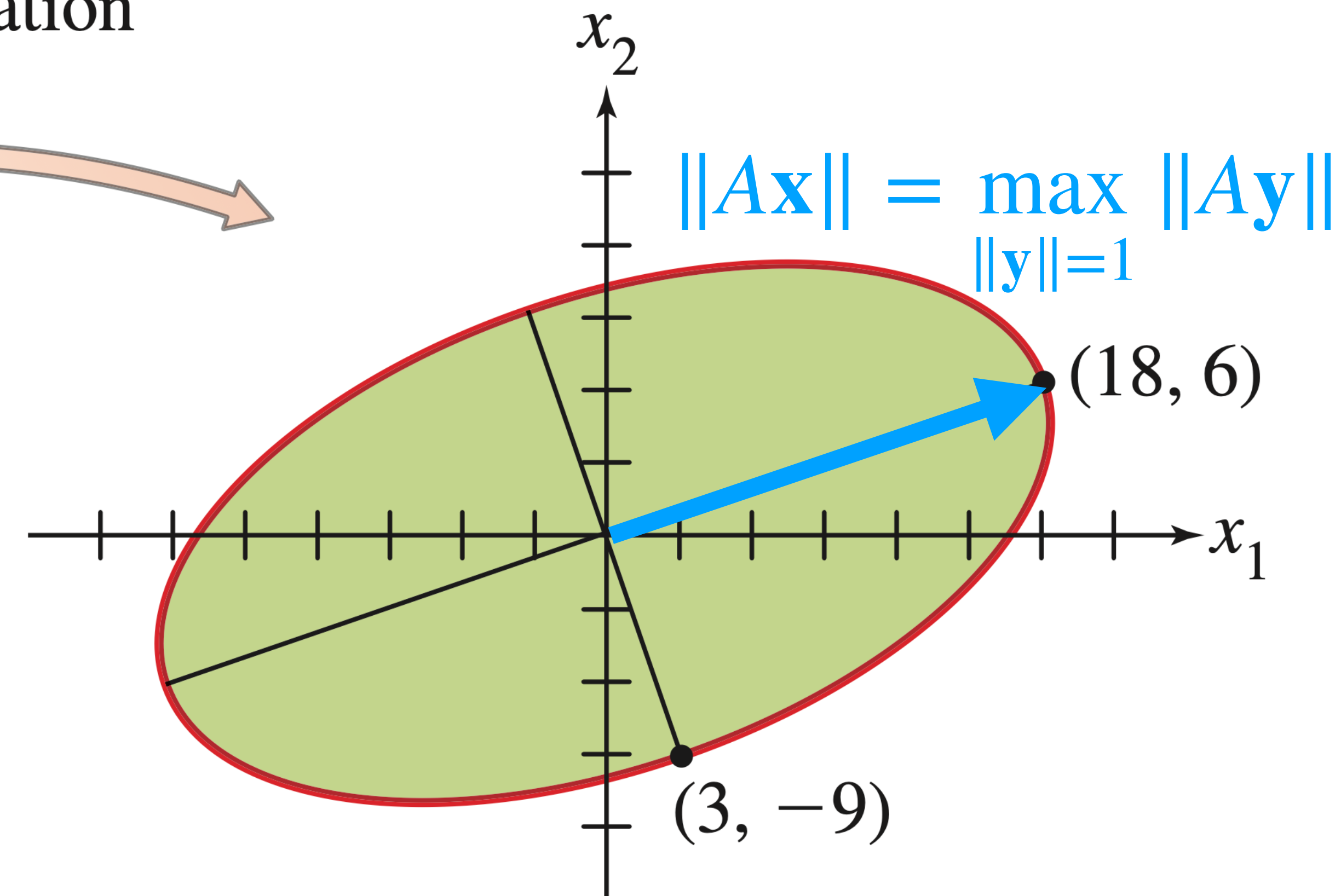
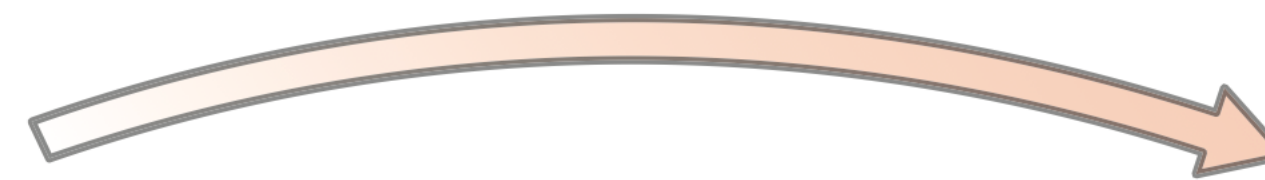


The Picture

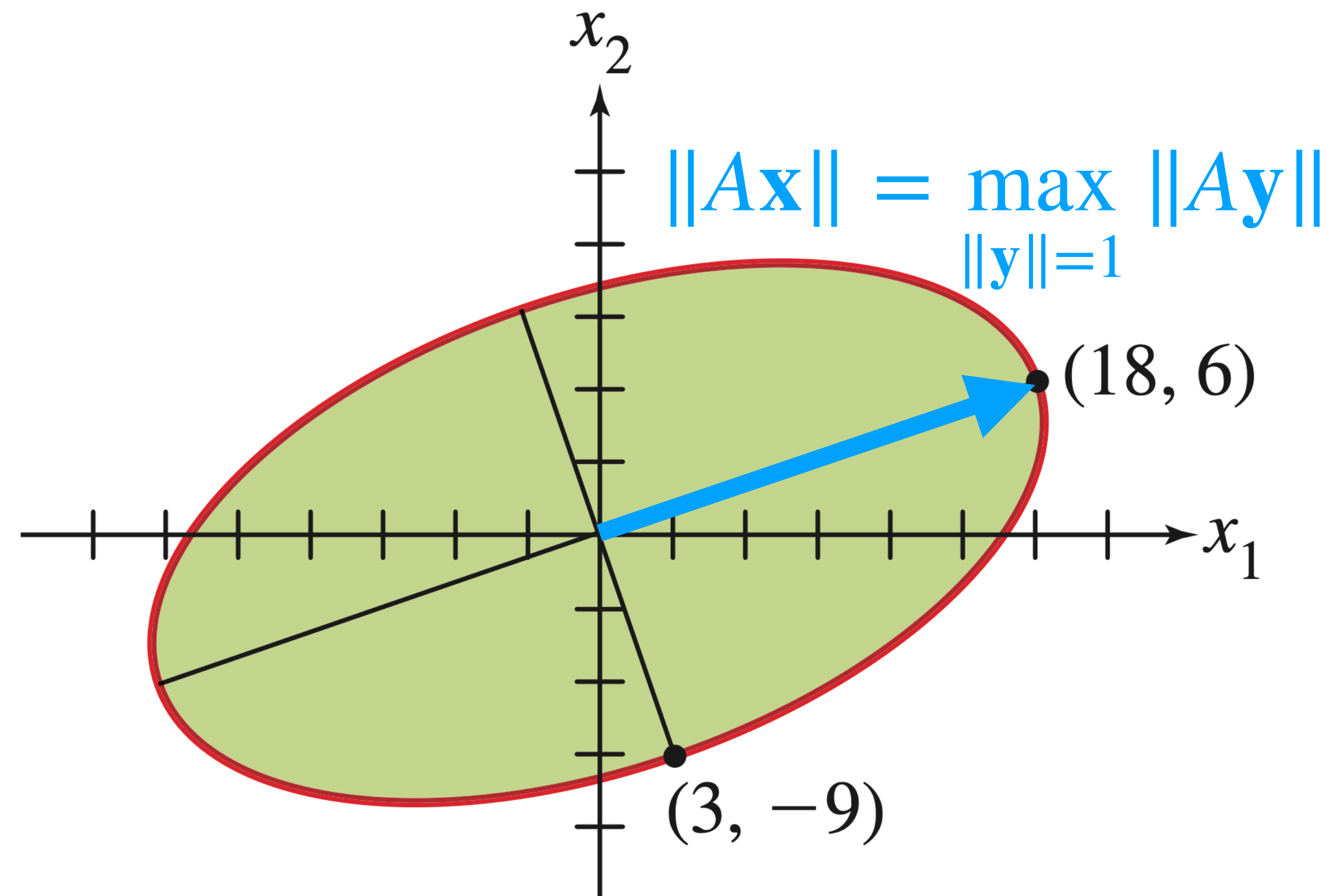
The longest end of the ellipse is the solution to a constrained optimization problem



Multiplication
by A

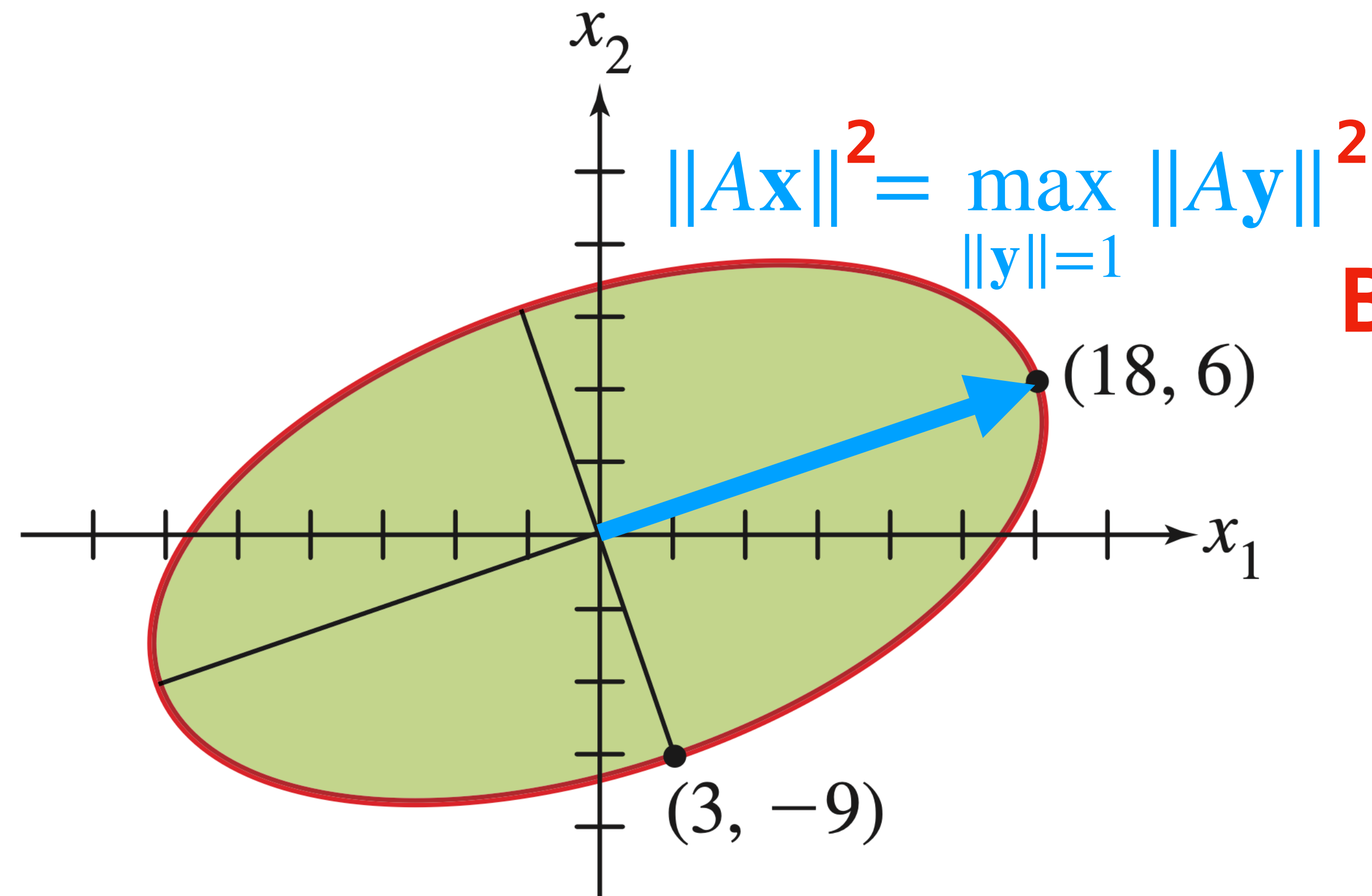


The Picture



This is not a quadratic form...

The Picture



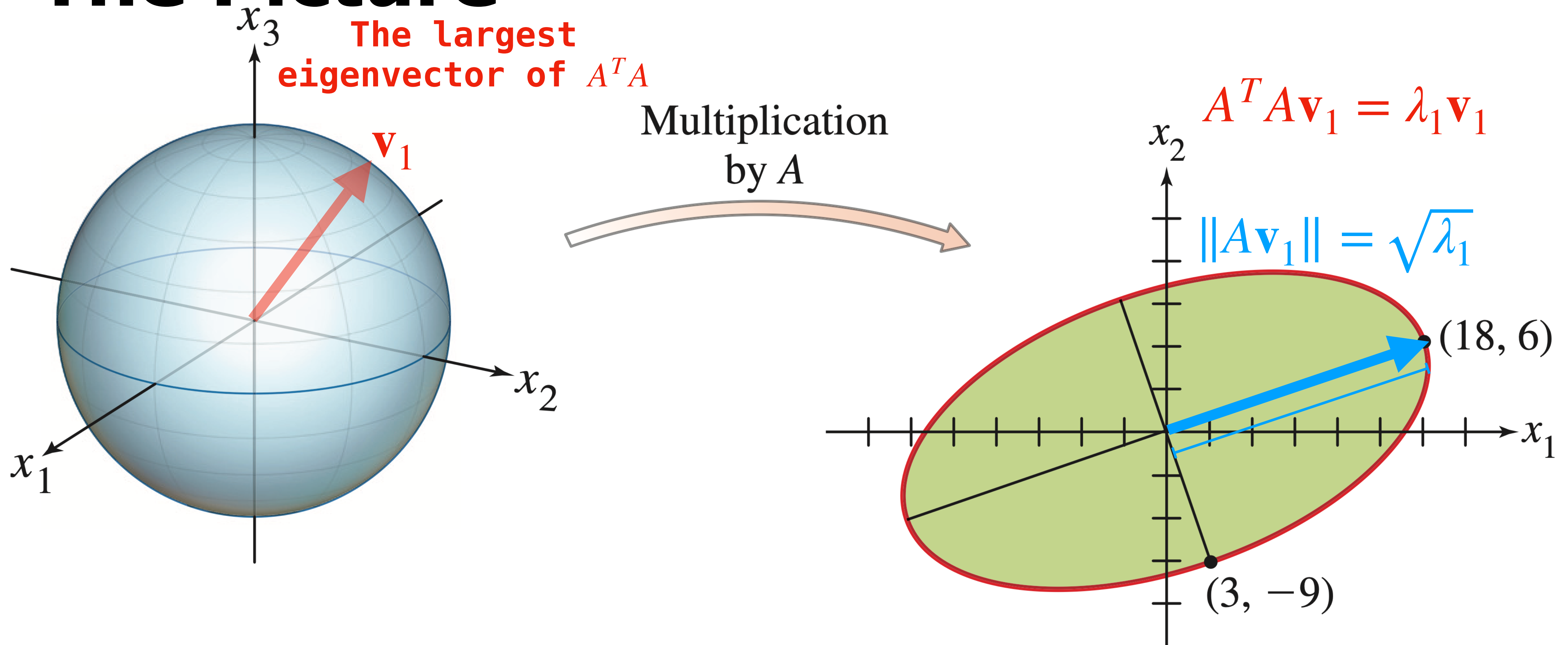
But this is

This is not a quadratic form...

A Quadratic Form

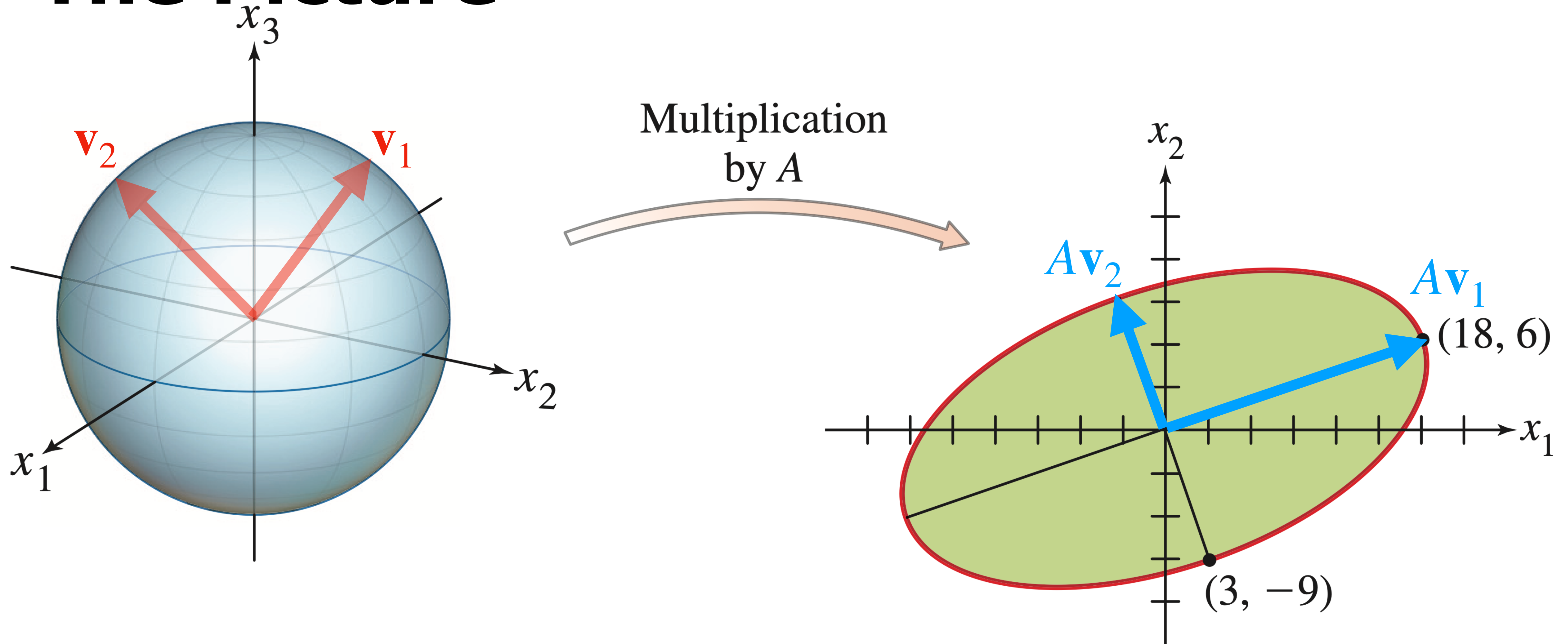
What does $\|A\mathbf{x}\|^2$ look like?

The Picture



$\mathbf{v}_1$ solves the constrained optimization problem

The Picture



The second eigenvector is sent to the *minimum* principle axis

Properties of $A^T A$

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It's symmetric

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So its *orthogonally diagonalizable*

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There is an orthogonal basis of eigenvectors

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It's eigenvalues are nonnegative

Properties of $A^T A$

It's symmetric

So its *orthogonally diagonalizable*

There is an orthogonal basis of eigenvectors

It's eigenvalues are nonnegative

It's positive semidefinite

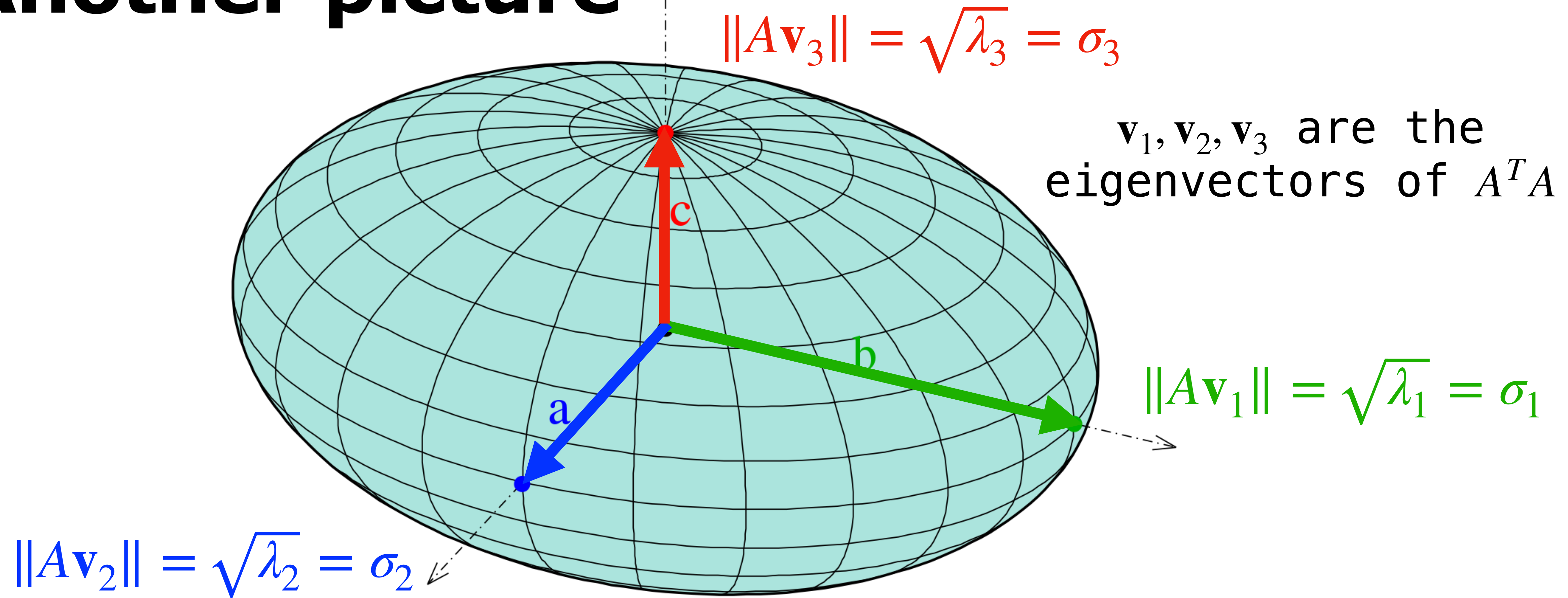
Singular Values

Def. For $A \in \mathbb{R}^{m \times n}$ the **singular values** of A are the n values

$$\sigma_1 \geq \sigma_2 \dots \geq \sigma_n \geq 0$$

where $\sigma_i = \sqrt{\lambda_i}$ and λ_i is an eigenvalue of $A^T A$

Another picture



The **singular values** are the lengths of the *axes of symmetry* of the ellipsoid after transforming the unit sphere

Every $m \times n$ matrix transforms the
unit m -sphere into an n -ellipsoid

So every $m \times n$ matrix has n
singular values

What else can we say?

Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be an **orthogonal** eigenbasis of $\mathbb{R}^n$ for $A^T A$ and suppose A has r *nonzero* singular values

Thm. $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ is an orthogonal basis of $\text{Col}(A)$

What else can we say?

Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be an **orthogonal** eigenbasis of $\mathbb{R}^n$ for $A^T A$ and suppose A has r *nonzero* singular values

Thm. $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ is an orthogonal basis of $\text{Col}(A)$

This is the most important theorem for SVD

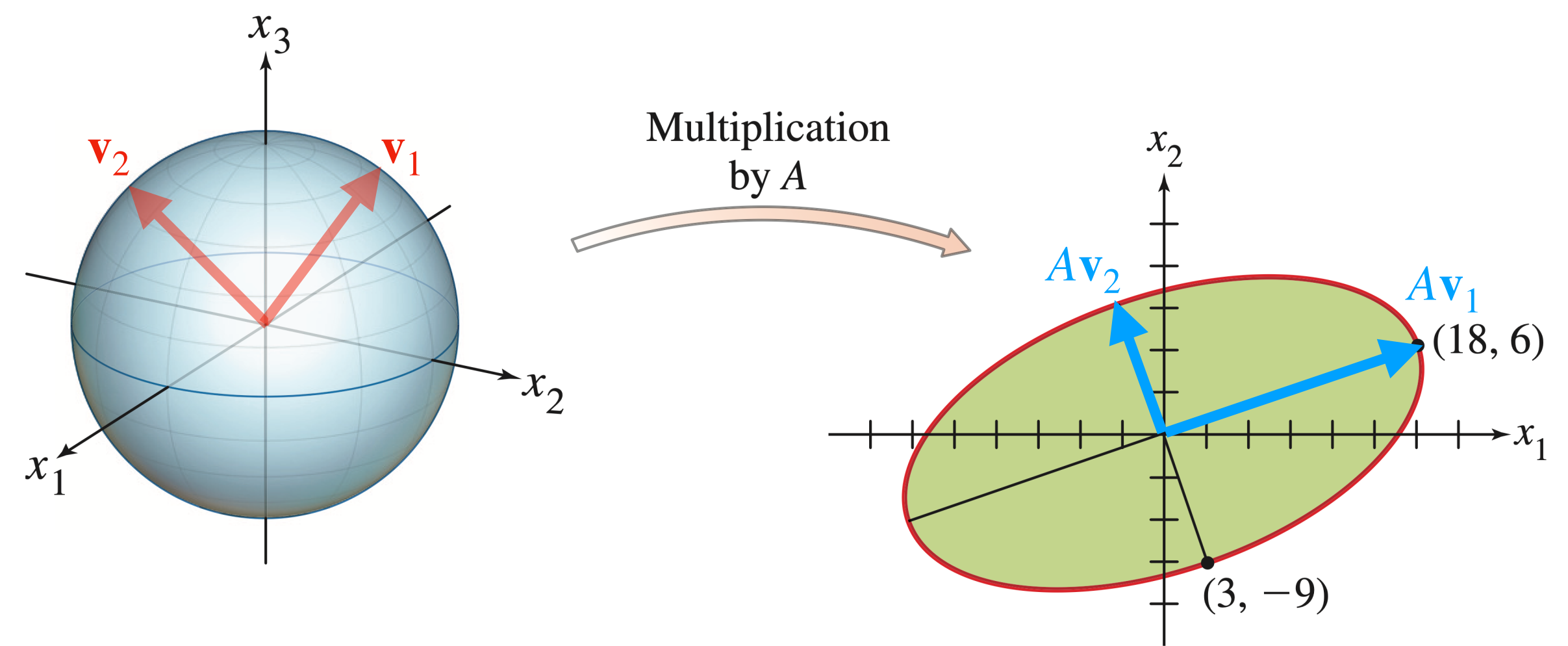
Verifying it

Let's show $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ are orthogonal (and L.I.)

Verifying it

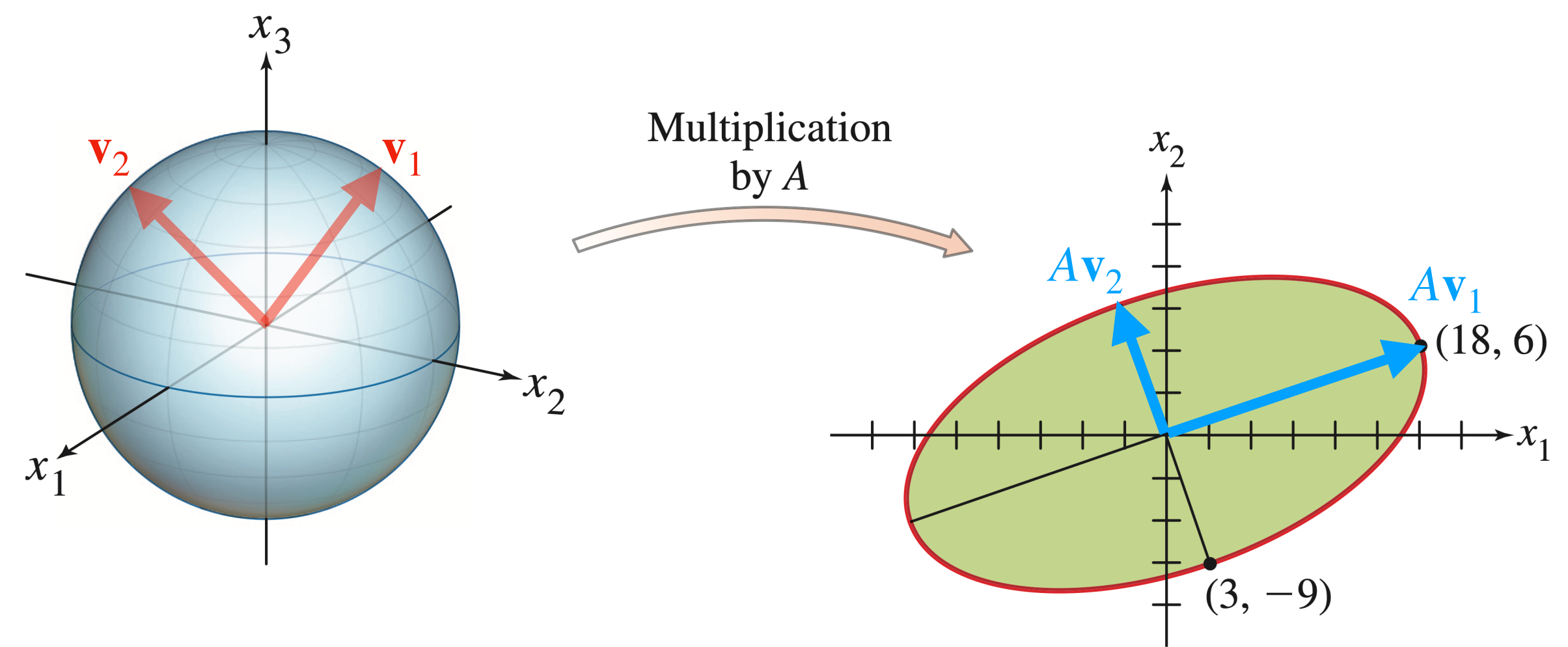
Let's show $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ span $\text{Col}(A)$

Putting it together



Putting it together

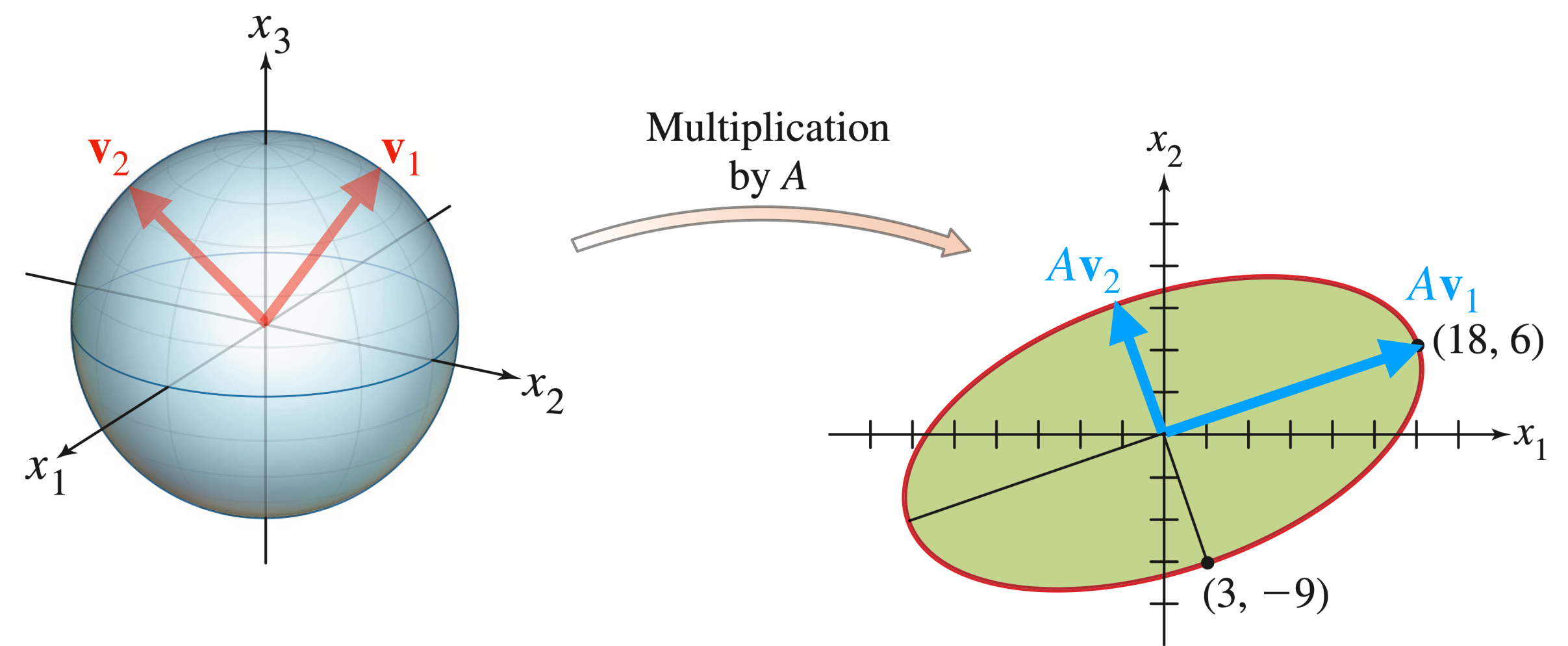
Let $A \in \mathbb{R}^{m \times n}$ be of rank r



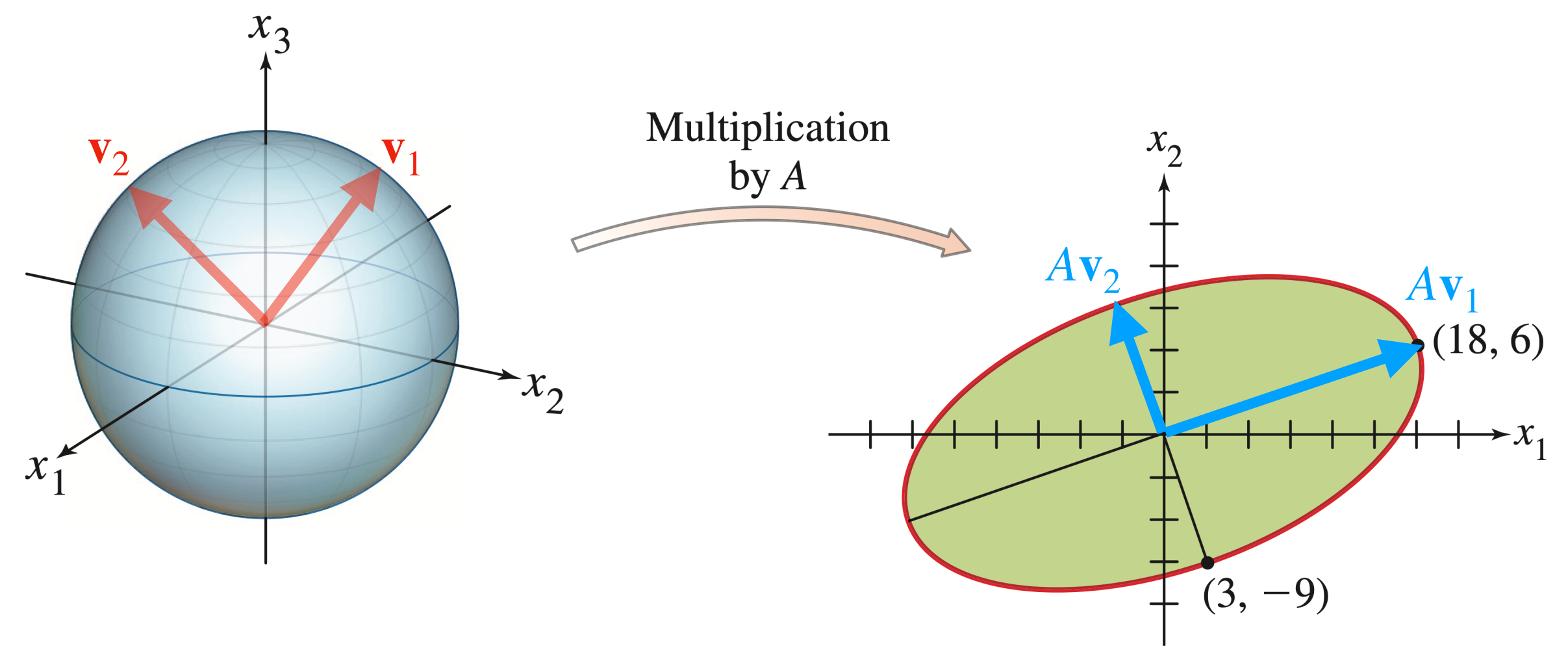
Putting it together

Let $A \in \mathbb{R}^{m \times n}$ be of rank r

What we know:



Putting it together

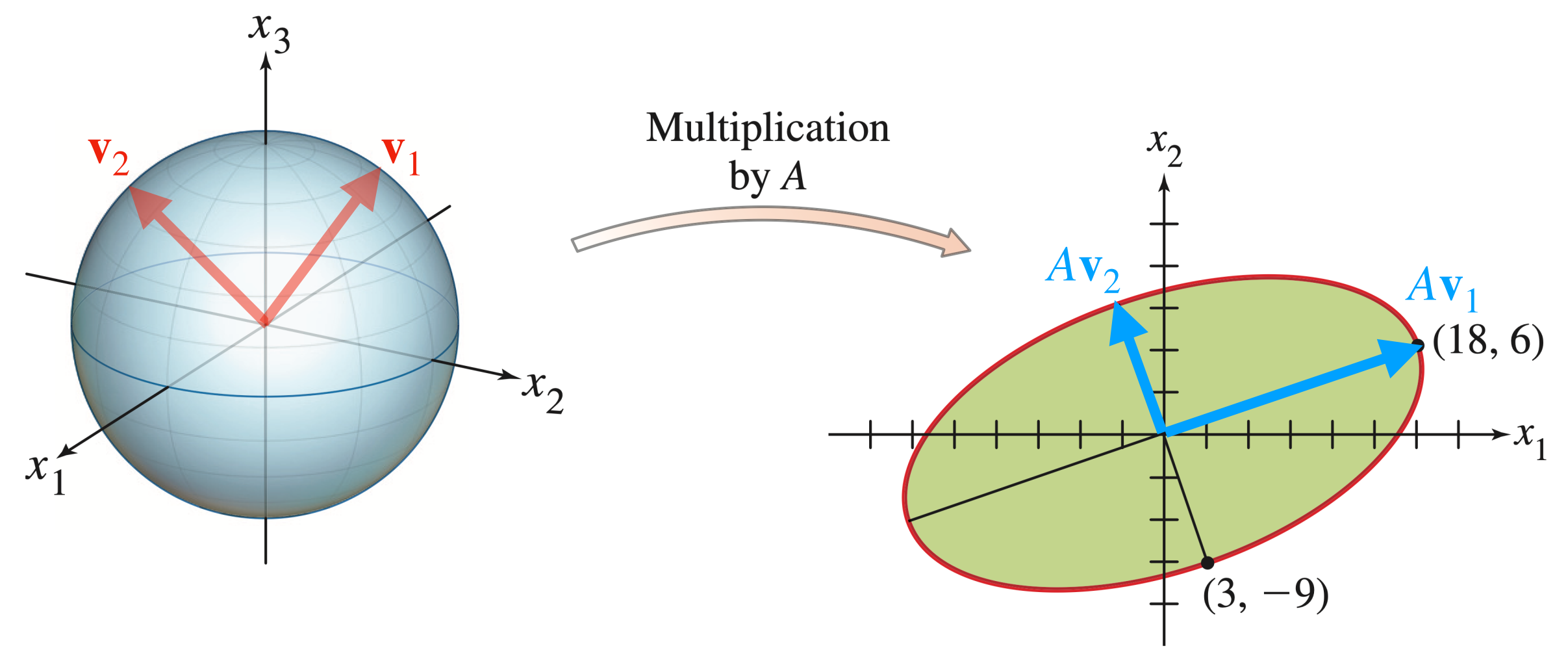


Let $A \in \mathbb{R}^{m \times n}$ be of rank r

What we know:

» We can find orthonormal vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ in $\mathbb{R}^n$ such that $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ in $\mathbb{R}^m$ form an orthogonal basis for $\text{Col}(A)$

Putting it together



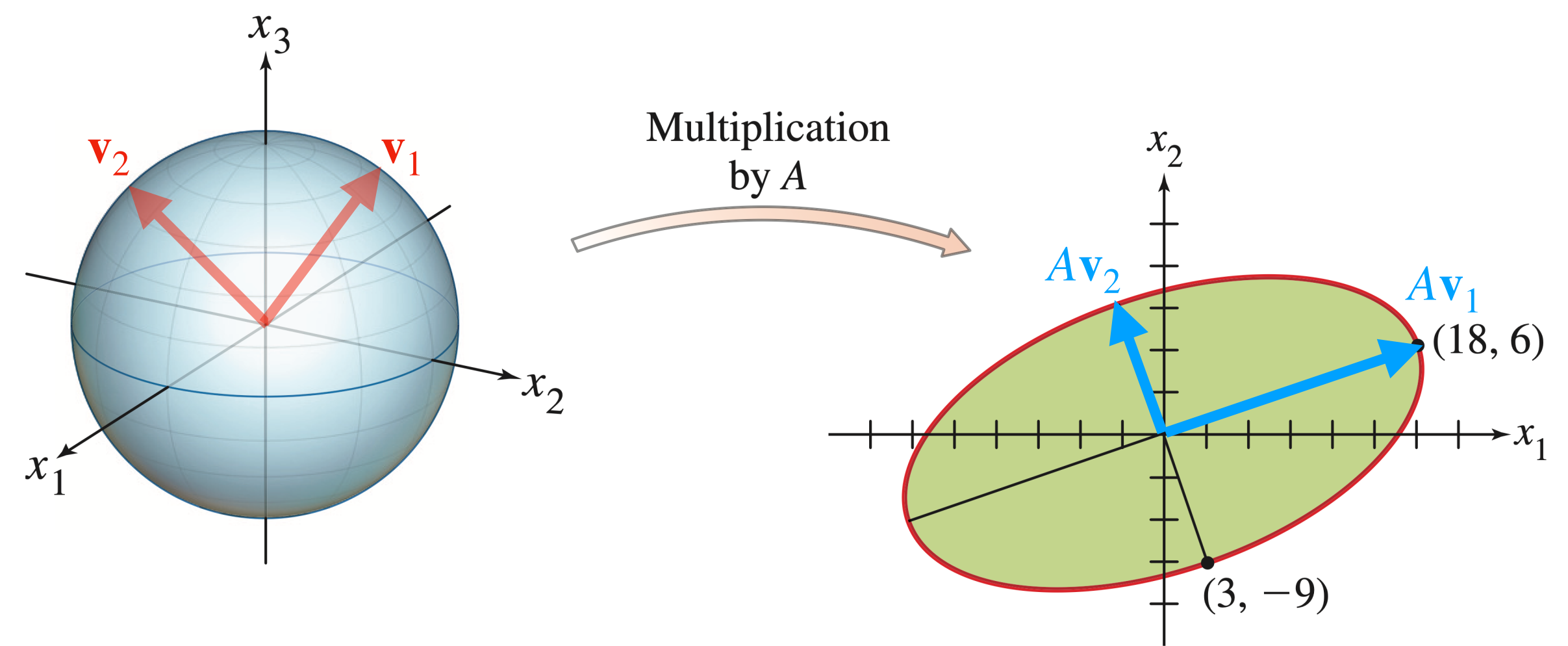
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» So if we take $\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|}$, we get an **orthonormal** basis of $\text{Col}(A)$

Putting it together



Let $A \in \mathbb{R}^{m \times n}$ be of rank r

What we know:

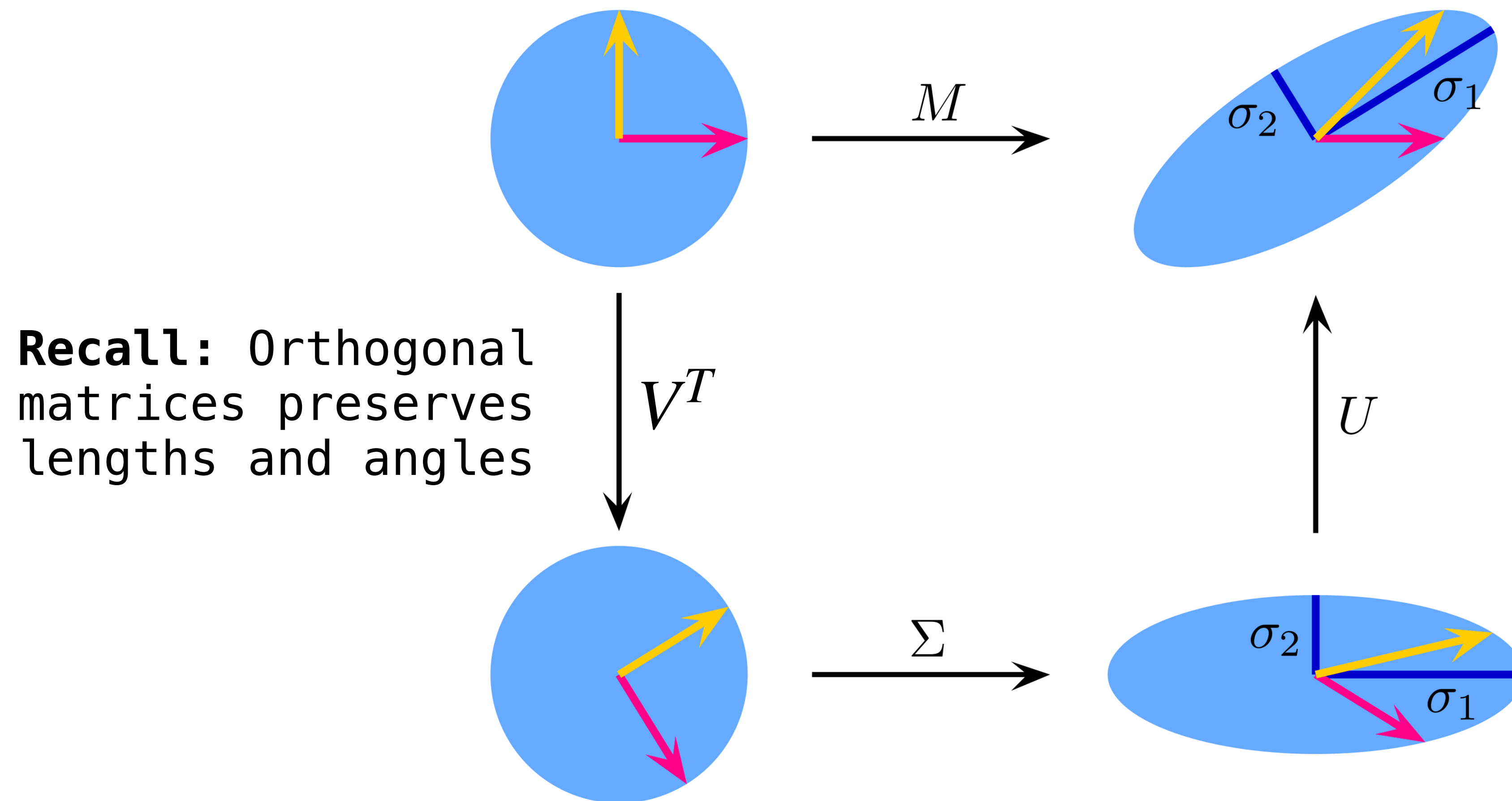
» We can find orthonormal vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ in $\mathbb{R}^n$ such that $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ in $\mathbb{R}^m$ form an orthogonal basis for $\text{Col}(A)$

» So if we take $\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|}$, we get an **orthonormal** basis of $\text{Col}(A)$

» And we can extend this to $\mathbf{u}_1, \dots, \mathbf{u}_m$ an orthonormal basis of $\mathbb{R}^m$ (via Gram-Schmidt)

Singular Value Decomposition

High Level View of the Decomposition



$$M = U \cdot \Sigma \cdot V^T$$

The Important Equality

$$\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|}$$

$$A\mathbf{v}_i = \|A\mathbf{v}_i\|\mathbf{u}_i = \sigma_i\mathbf{u}_i$$

The Important Equality

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Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value,
which is the length $\|A\mathbf{v}_i\|$

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Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value,
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What happens when we write this in matrix form?

The Important Equality

$$A[\mathbf{v}_1 \ \dots \ \mathbf{v}_n] = [\sigma_1 \mathbf{u}_1 \ \dots \ \sigma_n \mathbf{u}_n]$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value,
which is the length $\|A\mathbf{v}_i\|$

The Important Equality

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Let's take $V = [\mathbf{v}_1 \ \dots \ \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \ \dots \ \mathbf{u}_m]$ and

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Let's take $V = [\mathbf{v}_1 \ \dots \ \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \ \dots \ \mathbf{u}_m]$ and

$$\Sigma = \begin{matrix} m > n \\ \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \\ 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} \end{matrix} \text{ or } \Sigma = \begin{matrix} m < n \\ \begin{bmatrix} \sigma_1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_m & 0 & \dots & 0 \end{bmatrix} \end{matrix} \text{ or } \Sigma = \begin{matrix} m = n \\ \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \end{bmatrix} \end{matrix}$$

The Important Equality

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remember: U is orthonormal

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The Important Equality

$$AV = U\Sigma$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value, which is the length $\|A\mathbf{v}_i\|$

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The Important Equality

$$\overset{m \times n}{A} \underset{n \times n}{V} = \overset{m \times m}{U} \underset{m \times n}{\Sigma}$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value, which is the length $\|A\mathbf{v}_i\|$

Let's take $V = [\mathbf{v}_1 \dots \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \dots \mathbf{u}_m]$ and

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The Important Equality

$$AVV^T = U\Sigma V^T$$

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The Important Equality

$$A = U\Sigma V^T$$

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The Important Equality

singular value decomposition

$$A = U \Sigma V^T$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value, which is the length $\|A \mathbf{v}_i\|$

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Singular Value Decomposition

Thm. For $A \in \mathbb{R}^{m \times n}$, there are *orthogonal* matrices $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ such that

$$A = \overset{m \times m}{\boxed{U}} \underset{m \times n}{\Sigma} \overset{n \times n}{\boxed{V^T}}$$

where diagonal entries* of Σ are $\sigma_1, \dots, \sigma_n$ the singular values of A

* these are diagonal entries in a *non-square* matrix

Singular Value Decomposition

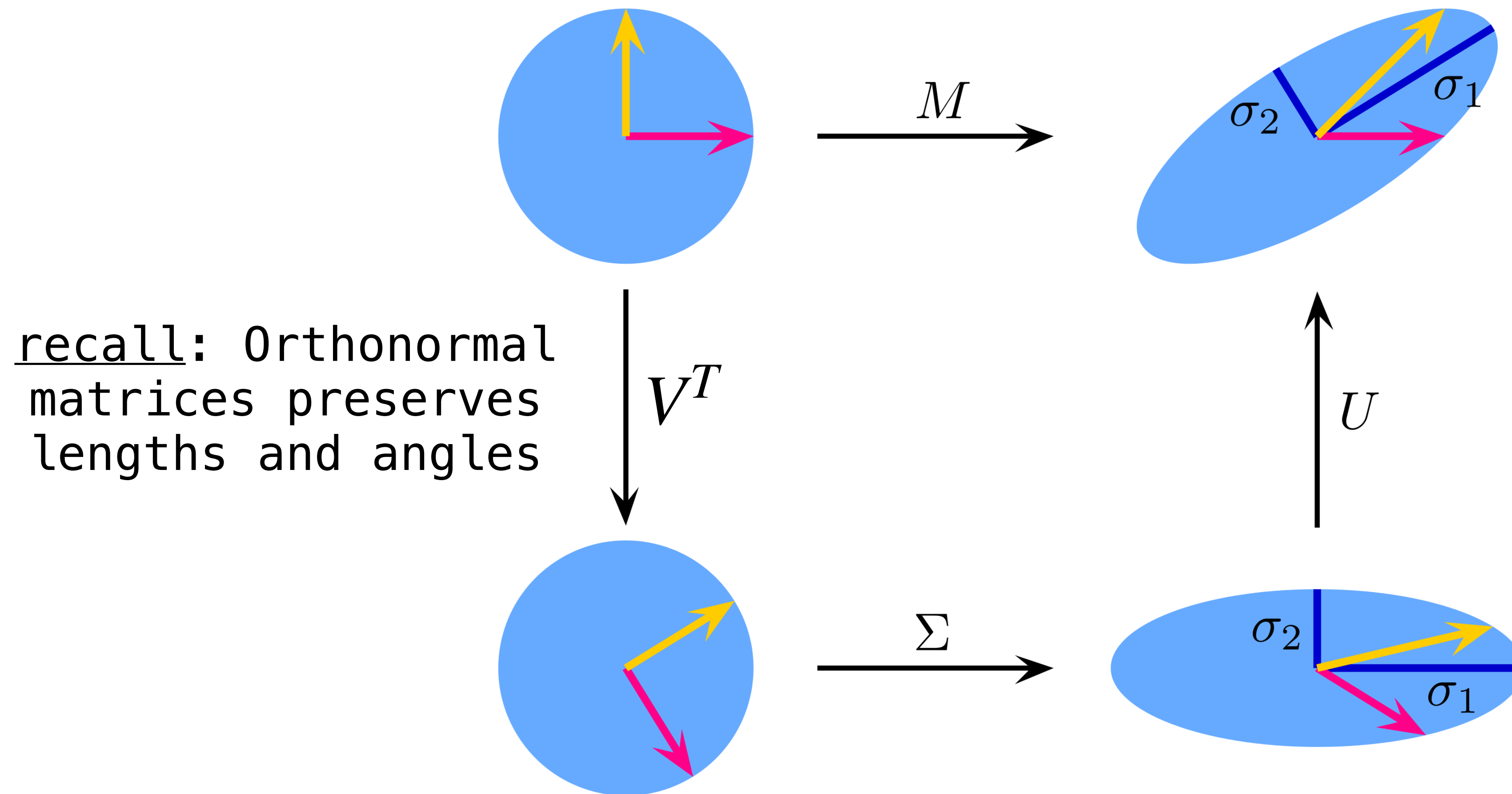
Thm. For $A \in \mathbb{R}^{m \times n}$, there are *orthogonal* matrices
 $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ such that
left singular *right singular*

$$A = \overset{m \times m}{U} \overset{m \times n}{\Sigma} \overset{n \times n}{V^T}$$

where diagonal entries* of Σ are $\sigma_1, \dots, \sigma_n$ the singular values of A

* these are diagonal entries in a *non-square* matrix

The Picture (Again)



$$M = U \cdot \Sigma \cdot V^T$$

Finding a SVD

Step 1: Set up Σ

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

The **singular values** are the square roots of the eigenvalues of $A^T A$ (or AA^T)

Step 2: Set up V

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

Find an orthonormal eigenbasis for $A^T A$

Step 3: Set up U (Part 1)

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

If $\mathbf{v}_1, \dots, \mathbf{v}_n$ is an eigenbasis of $\mathbb{R}^n$ (in decreasing order of eigenvalue), then $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ is an eigenbasis of $\text{Col}(A)$ (where r is the rank of A). These vectors can be normalized and made the first r columns of U

Step 4: Set up U (Part 2)

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

If $m > r$, then extend $\mathbf{u}_1, \dots, \mathbf{u}_r$ until it has m orthonormal vectors

Step 5: Put it all together

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

SVD in NumPy

In reality, we will almost never build SVDs by hand. We can use:

`numpy.linalg.svd`

Let's do a quick demo...

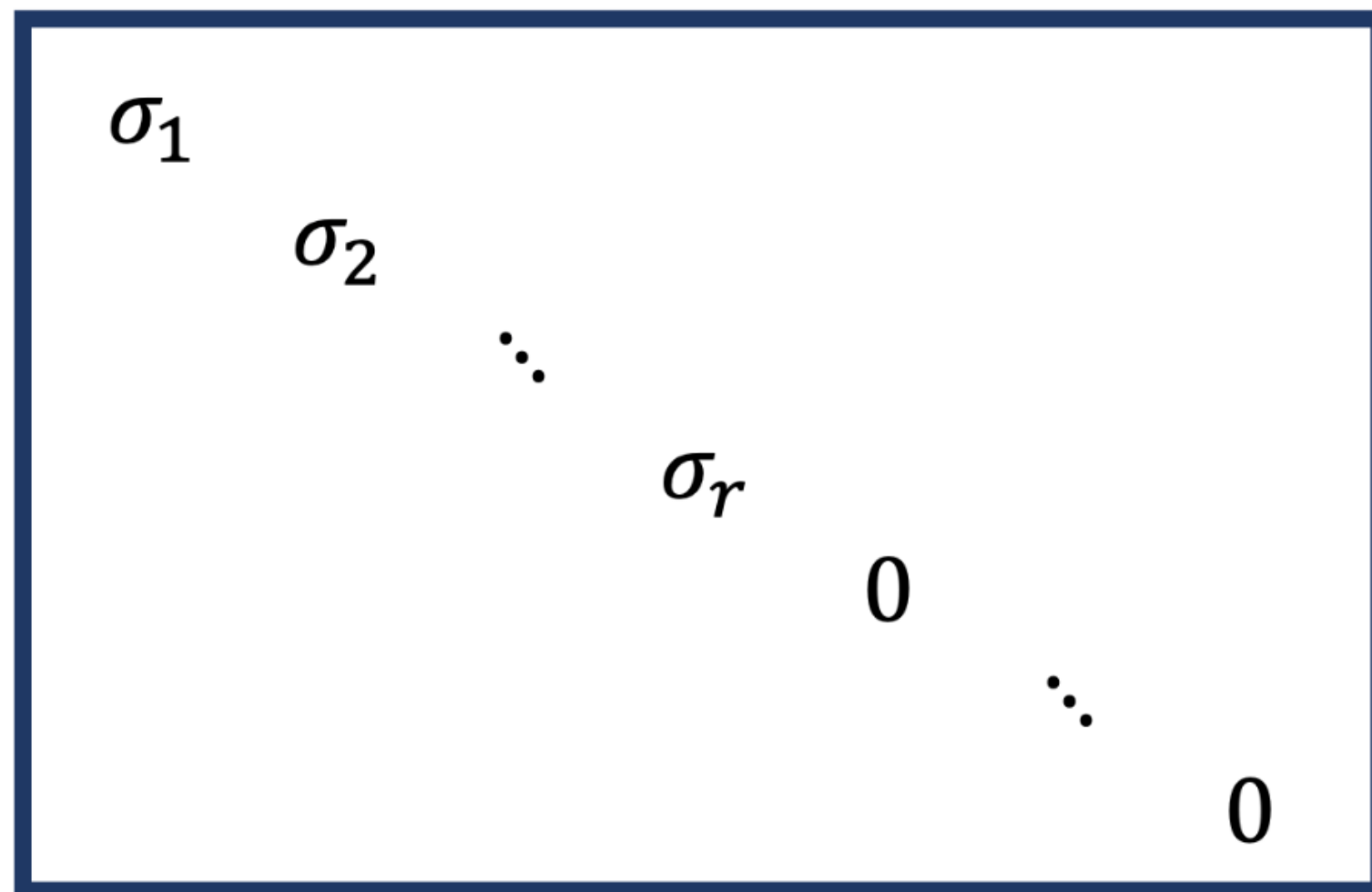
Pseudoinverses

SVD (The Picture)

U



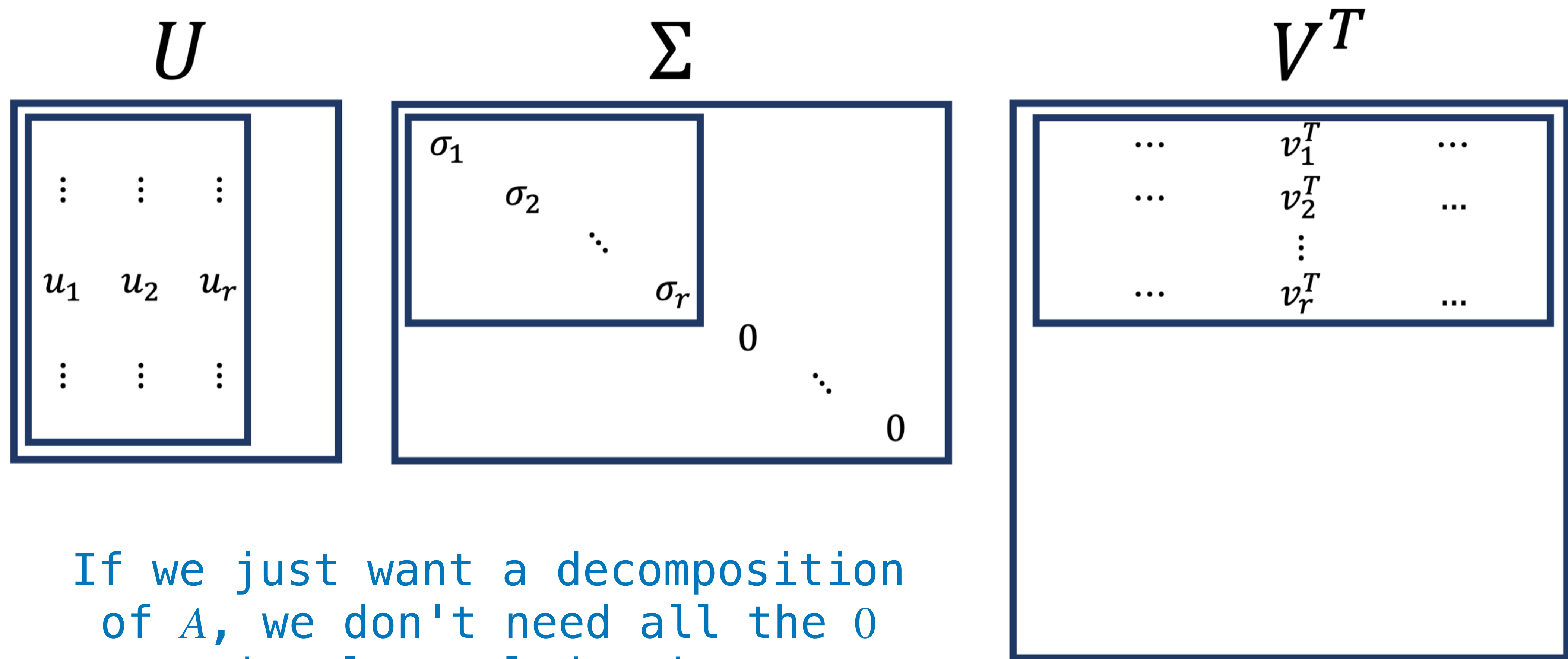
Σ



V^T



Reduced SVD (The Picture)



If we just want a decomposition of A , we don't need all the 0 singular values in Σ

The Reduced SVD

Thm. For every matrix A of rank r , there is an orthonormal matrix $U \in \mathbb{R}^{m \times r}$, a diagonal matrix $\Sigma \in \mathbb{R}^{r \times r}$ with **positive** entries on the diagonal, and an orthonormal matrix $V \in \mathbb{R}^{n \times r}$ such that

$$A = U\Sigma V^T$$

Back to the Example

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

The Pseudoinverse

Def. Given a reduced SVD $A = U\Sigma V^T$, the **pseudoinverse** of A is $A^+ = V\Sigma^{-1}U^T$

Thm. $A^+\mathbf{b}$ is the *minimum length* **least squares solution** of $A\mathbf{x} = \mathbf{b}$

(in Python we have *`numpy.linalg.pinv`*)

Recall: Least Squares in NumPy

numpy.linalg.lstsq

`linalg.lstsq(a, b, rcond='warn')`

[\[source\]](#)

Return the least-squares solution to a linear matrix equation.

Computes the vector x that approximately solves the equation $a @ x = b$. The equation may be under-, well-, or over-determined (i.e., the number of linearly independent rows of a can be less than, equal to, or greater than its number of linearly independent columns). If a is square and of full rank, then x (but for round-off error) is the “exact” solution of the equation. Else, x minimizes the Euclidean 2-norm $||b - ax||$. If there are multiple minimizing solutions, the one with the smallest 2-norm $||x||$ is returned.

Parameters: a : (M, N) *array_like*

“Coefficient” matrix.

b : $\{(M,), (M, K)\}$ *array_like*

Ordinate or “dependent variable” values. If b is two-dimensional, the least-squares solution is calculated for each of the K columns of b .

$rcond$: *float. optional*

Recall: Least Squares in NumPy

numpy.linalg.lstsq

`linalg.lstsq(a, b, rcond='warn')`

[\[source\]](#)

Return the least-squares solution to a linear matrix equation.

Computes the vector x that approximately solves the equation $a @ x = b$. The equation may be under-, well-, or over-determined (i.e., the number of linearly independent rows of a can be less than, equal to, or greater than its number of linearly independent columns). If a is square and of full rank, then x (but for round-off error) is the “exact” solution of the equation. Else, x minimizes the Euclidean 2-norm $||b - ax||$. If there are multiple minimizing solutions, the one with the smallest 2-norm $||x||$ is returned.

NumPy chooses the shortest vector

Parameters: $a : (M, N)$ *array_like*

“Coefficient” matrix.

$b : \{(M,), (M, K)\}$ *array_like*

Ordinate or “dependent variable” values. If b is two-dimensional, the least-squares solution is calculated for each of the K columns of b .

$rcond : float$ *optional*

Recall: Least Squares in NumPy

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because they use SVD!